

Product Requirement Document (PRD)

Product Name

Research Document Q&A Assistant (Phase 1)

Product Vision

Enable users to upload research papers and interact with them through natural language questions, receiving contextually relevant and evidence-based answers.

Problem Statement

Researchers and students often need to manually search through lengthy documents to locate specific information. This process is time-consuming and inefficient.

The system should allow users to upload documents and ask questions in natural language to quickly retrieve relevant information.

Target Users

- Students
 - Researchers
 - Engineers
 - Technical professionals
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User Stories

US1

As a user, I want to upload PDF documents so that I can query them later.

US2

As a user, I want to ask questions about uploaded documents.

US3

As a user, I want answers grounded in the uploaded content.

US4

As a user, I want citations showing where information came from.

US5

As a user, I want responses within a few seconds.

Functional Requirements

FR1 Document Upload

The system shall allow uploading PDF files.

FR2 Document Processing

The system shall extract text from uploaded PDFs.

FR3 Chunk Generation

The system shall split documents into overlapping chunks.

FR4 Embedding Generation

The system shall generate vector embeddings for chunks.

FR5 Vector Storage

The system shall store embeddings inside a vector database.

FR6 Semantic Retrieval

The system shall retrieve top-k relevant chunks based on similarity.

FR7 Question Answering

The system shall generate answers using retrieved chunks.

FR8 Source Citation

The system shall return the source chunks used to answer.

FR9 Retrieval Agent

The system shall include a retrieval agent whose primary responsibility is to identify and retrieve the most relevant document chunks for a given user query.

FR11 Iterative Retrieval

The retrieval agent shall be capable of performing multiple retrieval steps when the initially retrieved documents are insufficient or incomplete.

FR12 Thought-and-Action Retrieval Strategy

The retrieval process shall follow a Thought-and-Action (ReAct) approach, where the agent reasons about the query, performs retrieval actions, evaluates the retrieved information, and decides whether additional retrieval steps are required.

Non-Functional Requirements

Accuracy

Answers should be based only on retrieved content.

Scalability

Support documents containing hundreds of pages.

Modularity

Embedding models and LLMs should be replaceable.

Latency

Response time should preferably be under 5 seconds.

Extensibility

Architecture should support future agentic capabilities.

Success Metrics

- Answer relevance
- Citation correctness
- Response latency
- Retrieval quality
- User satisfaction